

p1

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1.1

ASE3796 HW08

1. $Y = f(X) = X^3$ $X \sim N(1, \sigma^2)$; $\bar{X} = 1$, $\text{var}(X) = \sigma^2 = V$

$E[f(X)] = f(\bar{X}) = 1^3 = 1$

$\text{var}(f(X)) = \nabla f(\bar{X}) \nabla f(\bar{X})^T = 3\sigma^2 \cdot 3$

$\nabla f(\bar{X}) = 3\bar{X}^2 = 3$

$\text{var}(f(\bar{X})) = 9\sigma^2$

if $\sigma^2 = 0.01$, $\begin{cases} E(Y) = 1 \\ \text{var}(Y) = 9 \times 10^{-2} \end{cases}$

if $\sigma^2 = 0.1$, $\begin{cases} E(Y) = 1 \\ \text{var}(Y) = 9 \times 10^{-1} \end{cases}$

1.2

In [5]:

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
def f(sigma, M):

    X = np.random.normal(1, sigma, M)
    Y = X**3

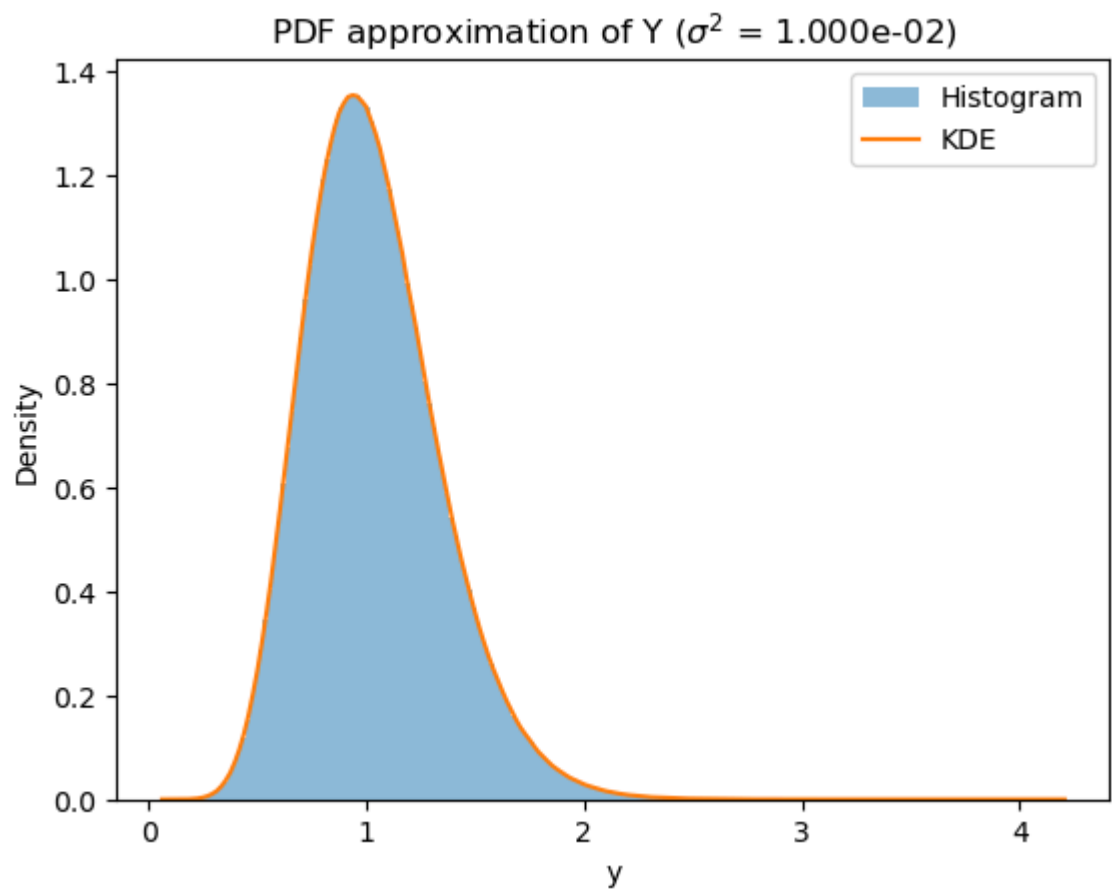
    print("Mean:", np.mean(Y))
    print("Variance:", np.var(Y))

    plt.figure()
    plt.hist(Y, bins=200, density=True, alpha=0.5, label="Histogram")
    sns.kdeplot(Y, label="KDE")

    plt.xlabel("y")
    plt.ylabel("Density")
    plt.legend()
    plt.title(f"PDF approximation of Y ( $\sigma^2 = \{{\sigma}^2:.3e\}$ ")
    plt.show()

M = 10_000_000
sigma2 = 0.01
sigma = np.sqrt(sigma2)
f(sigma, M)
```

```
Mean: 1.0300767966822306
Variance: 0.09366696825670462
```

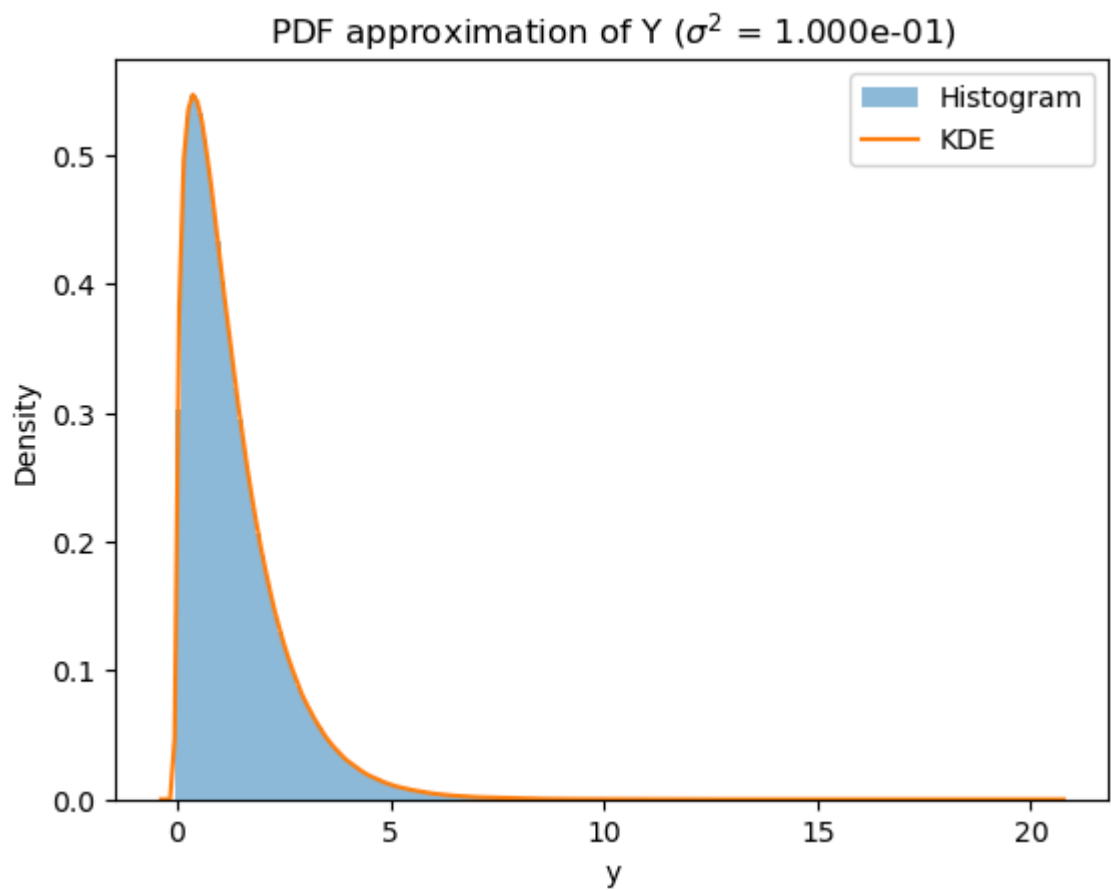


1.3

In [6]:

```
M = 10_000_000
sigma2 = 0.1
sigma = np.sqrt(sigma2)
f(sigma, M)
```

```
Mean: 1.2998122444060864
Variance: 1.2750207499557036
```



1.4

The perturbation method provides a good approximation when the input variance is small. For $\sigma^2 = 0.01$, the Monte Carlo estimates of the mean and variance are close to the first-order perturbation predictions, indicating that the linearization of the nonlinear function $f(X) = X^3$ around the mean is sufficiently accurate. However, for $\sigma^2 = 0.1$, the empirical mean and variance deviate significantly from the perturbation estimates. This occurs because the cubic transformation introduces strong nonlinear effects, and the first-order Taylor approximation is no longer adequate over the wider range of possible input values. This illustrates that perturbation methods are most reliable when the input uncertainty is small or when the function is approximately linear near the mean. For strongly nonlinear functions or larger uncertainties, higher-order expansions or sampling-based methods such as Monte Carlo simulation are required for accurate uncertainty propagation.